

Extreme tree growth responses...

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Tree-ring data offer a unique opportunity to learn how trees grow in a non-stationary climate with more frequent extreme events. However, disentangling tree responses to long-term trends and extreme events requires careful consideration of the complex interactions between climatic factors and size-dependent developmental stages. We developed a new modeling approach that integrates individual-level growth trends with species-specific climate sensitivities, and that is specifically designed to capture extreme growth responses across species and ecosystems. We applied it to tree-ring records from Western North America, from 1900 to today. Our model identifies the significant drought period of the 200X-200Y as causing exceptional reductions in tree growth, well beyond those predicted alone due to temperature, winter precipitation (driving soil moisture) and vapor pressure deficit, with some of the largest declines in XX species and YY regions. These results highlight that tree growth is a critical indicator of drought, with growth declines that may be underestimated in many current models.